

INDUSTRIAL SAFETY & COMMUNICATION

Visibility where the environment demands more.

HAZARDOUS-AREA CCTV | PABX EXPANSION ROADMAP



FLAMEPROOF SURVEILLANCE



PLANT COMMUNICATION

**Engineered by D-TECH for Alchemie Finechem - Block 11,
Saykha**

A documented hazardous-area CCTV deployment with a clear path toward integrated PABX and emergency communication.

A focused hazardous-area deployment.

The purchase order validates a compact, engineered surveillance package for Block 11 at the Saykha manufacturing facility.

5

BULLET CAMERAS

Matrix SATATYA CIBR50FL40CWP units for defined observation points.

5

FLP ENCLOSURES

Aluminium housings with mounting brackets and PESO certification.

5 + 5

CONDUITS AND BOXES

M25 flameproof flexible conduits plus IP66 junction boxes.

BLOCK 11

DEPLOYMENT ZONE

Alchemie Finechem, Saykha GIDC, Vagra, Bharuch.

PROJECT INTENT

Create reliable visual coverage in a demanding industrial area using protected cameras, certified enclosures, sealed termination points and stable fabricated mounting structures.

COMMUNICATION OPPORTUNITY

Extend the same safety-led engineering discipline to a plant PABX architecture with safe-zone call control, hazardous-area field stations, paging and emergency call paths.

Evidence basis

Purchase Work Order dated 17-04-2026. Commercial pricing is intentionally excluded from this case study.

Standard devices are not enough in hazardous areas.

Chemical manufacturing zones require surveillance equipment and installation methods selected for safety, environmental exposure and operational continuity.

HAZARDOUS ATMOSPHERE

Field equipment must be protected through an appropriate certified enclosure strategy rather than exposed as standard commercial hardware.

DUST, MOISTURE & CORROSION

Cable entries, junction points and mounting interfaces need sealing and material choices suitable for industrial conditions.

SAFE CABLE TRANSITION

Field cabling requires disciplined routing through flameproof flexible conduits and weatherproof termination points.

RELIABLE CAMERA POSITIONING

Purpose-built stands and brackets are needed to achieve stable views without compromising the protected assembly.

OPERATIONS RESPONSE

Visibility is most valuable when security, maintenance and plant teams can coordinate quickly during an event.

FUTURE COMMUNICATION

The plant benefits from a roadmap that can connect field telephony, PABX, paging and emergency communication.

THE DESIGN QUESTION

How can the plant gain dependable hazardous-area visibility today while creating a structured foundation for faster voice and emergency coordination tomorrow?

Assess the zone. Engineer the assembly. Verify the path.

D-TECH treats every field point as a complete safety and performance assembly, not as an isolated camera or telephone.



D-TECH DELIVERY PRINCIPLE

The field point is successful only when safety, visibility, serviceability and operational ownership work together.

Two coordinated layers for safer plant operations.

The delivered CCTV layer creates hazardous-area visibility; the communication layer provides a scalable roadmap for voice, paging and emergency response.

DELIVERED CCTV LAYER

01

CAMERA

Matrix bullet camera

02

PROTECTION

PESO-certified FLP enclosure

03

FIELD PATH

M25 conduit + IP66 box

04

SAFE ZONE

Network / NVR / monitoring

PABX EXPANSION ROADMAP

01

FIELD CALL

Hazard-rated phone / station

02

CALL CONTROL

PABX in safe equipment room

03

BROADCAST

Paging / PA integration

04

RESPONSE

Security and emergency teams

ONE OPERATIONAL VIEW

Surveillance confirms what is happening. Plant communication helps the right team respond with speed and control.

A complete five-point field package.

The order combines cameras with the certified mechanical and cable-protection elements required for a disciplined hazardous-area installation.

5 NOS

Matrix SATATYA CIBR50FL40CWP bullet cameras

One-year warranty referenced in the purchase order.

5 NOS

Flameproof aluminium enclosures with mounting brackets

PESO-certified; one-year warranty referenced.

5 NOS

Flameproof flexible metallic conduits - M25

Protected cable transition at the field assembly.

5 NOS

Flameproof / weatherproof IP66 junction boxes

Sealed field termination and cable management.

5 NOS

Fabricated stands - 2 feet

Purpose-built mounting support for selected points.

5 NOS

Fabricated stands - 1 foot

Alternate mounting height for location-specific needs.

ORDER REFERENCE
Work Order 12/26-27/Fact/FLP CAMERA | Dated 17-04-2026 | Cost Center: Block No. 11

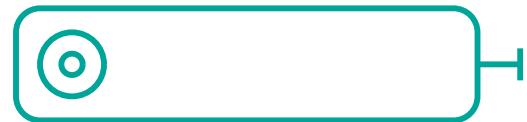
Every supplied product, shown as a field sketch.

Conceptual line drawings communicate the role and relationship of each ordered item. Final dimensions remain subject to approved product drawings and site conditions.



MATRIX BULLET CAMERA

5 NOS | Visual capture device



FLAMEPROOF ENCLOSURE

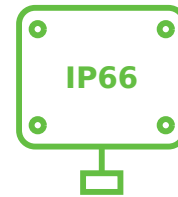
5 NOS | Aluminium, mounting bracket



M25 FLAMEPROOF FLEXIBLE CONDUIT

M25 FLEXIBLE CONDUIT

5 NOS | Protected cable transition



IP66 JUNCTION BOX

5 NOS | Sealed field termination



FABRICATED STAND

5 NOS | 2-foot configuration



FABRICATED STAND

5 NOS | 1-foot configuration

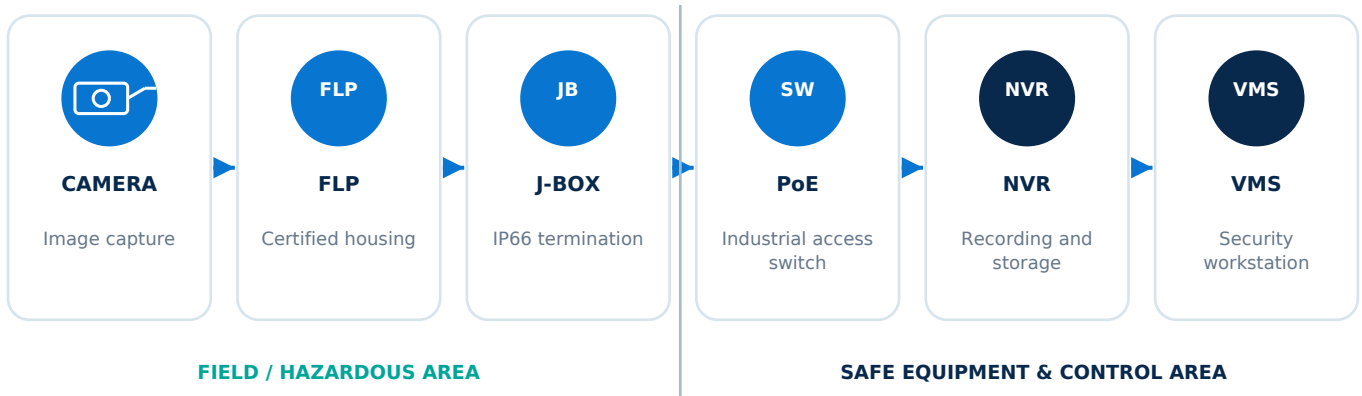
DRAWING NOTE

These sketches explain system intent and product function. Installation teams must follow OEM drawings, approved GA details and hazardous-area requirements.

Field devices connected to safe-zone operations.

A visual end-to-end design shows the delivered CCTV path and the separately scoped PABX communication path.

DELIVERED HAZARDOUS-AREA CCTV PATH



PABX / EMERGENCY COMMUNICATION PATH



CONNECTIVITY LEGEND

Delivered CCTV data path

PABX path - survey and separate BOQ required

Fiber or copper uplink selection depends on distance, bandwidth, pathway and plant network standards.

Protection is designed as a system.

The integrity of a hazardous-area camera point depends on every interface - housing, entry, termination, mounting and maintenance access.

01

CERTIFIED ENCLOSURE

PESO certification is documented for the supplied flameproof aluminium enclosure. Final suitability remains linked to the approved zone, gas group and installation practice.

02

SEALED TERMINATION

The IP66 junction box supports protection against dust and water ingress at the field termination point.

03

FLEXIBLE FIELD ENTRY

M25 flameproof flexible conduit manages the transition into the protected camera assembly while supporting routing discipline.

04

STABLE MOUNTING

One-foot and two-foot fabricated stands allow the camera position to be adapted to the location and required field of view.

05

SAFE-ZONE HANDOFF

Network, recording and PABX call-control equipment should remain in suitable safe areas unless independently certified for the installed zone.

06

MAINTAINABLE DESIGN

Labels, accessible connections, documented views and controlled permissions help sustain the solution after commissioning.

IMPORTANT ENGINEERING CONTROL

Hazardous-area classification, equipment certification, earthing, glands, cable entries and installation methods must be validated against the plant-approved design and applicable statutory requirements before commissioning.

From visibility to coordinated response.

A phased plant communication design can complement the CCTV deployment without overstating the current purchase-order scope.



SCOPE CLARITY

The supplied purchase order substantiates the hazardous-area CCTV package only. PABX equipment, quantities and commissioning should be confirmed through a separate survey, BOQ and approved order.

CASE STUDY CONCLUSION

Safer visibility today. Connected response tomorrow.

PROTECTED OBSERVATION

Five camera points engineered with certified enclosure protection for a demanding industrial area.

STRUCTURED FIELD DESIGN

Conduit, IP66 termination and fabricated stands form one maintainable field assembly.

OPERATIONAL CONFIDENCE

A documented solution supports security, plant administration and maintenance ownership.

SCALABLE COMMUNICATION

A clear PABX roadmap can connect field calls, paging, security and emergency response.

D-TECH: ONE PARTNER FOR FIELD SAFETY, CCTV, NETWORKING AND PLANT COMMUNICATION

D-TECH Solution Integrators Pvt. Ltd.

End-to-End IT Infrastructure, SAP, Custom Software & Automation Solutions

sales@dtechindia.com | www.dtechindia.com | +91 91574 20663